Unstrctured Mesh Generation
Technical Progress Report (Week 7)

Pranjal Kumar Shukla

FOSSEE Internship – Progress Report

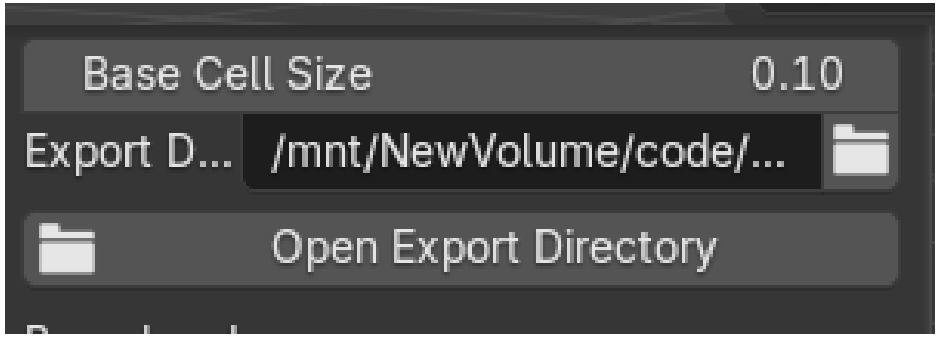
SCAI School

VIT Bhopal University

April 19, 2026

1. System File Explorer Integration

- **Objective**: Let users instantly browse their generated OpenFOAM dictionaries to verify configurations without using terminal commands.
- **Implementation**:
  - Added an **Open Export Directory** button under the Mesh Generation panel.
  - Leverages Python's `subprocess` API with `xdg-open` (Linux) to interact natively with the OS file explorer.
  - Automatically redirects users to exactly where the `system`, `constant`, and `0` folders are exported.



*The "Open Export Directory" button in the Workspace UI

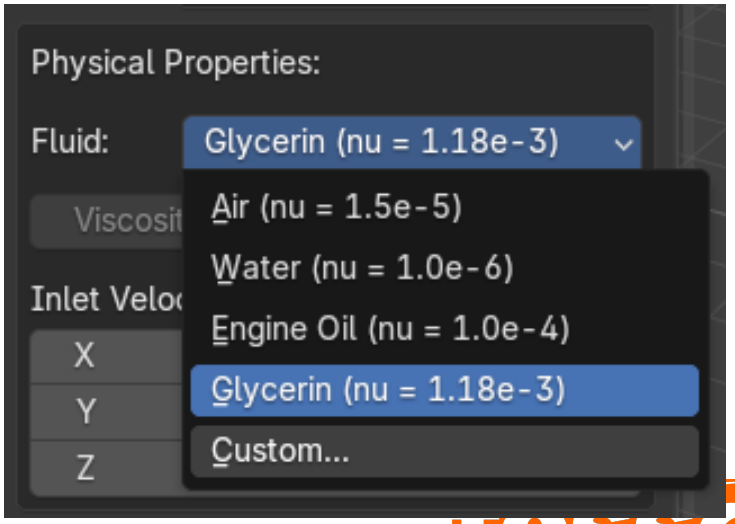
2. Custom Fluid Selection & Preset Mapping

Week 6 Limitation

Users had to manually calculate and input kinematic viscosity for specific fluids, adding friction to setup.

Week 7 Solution:

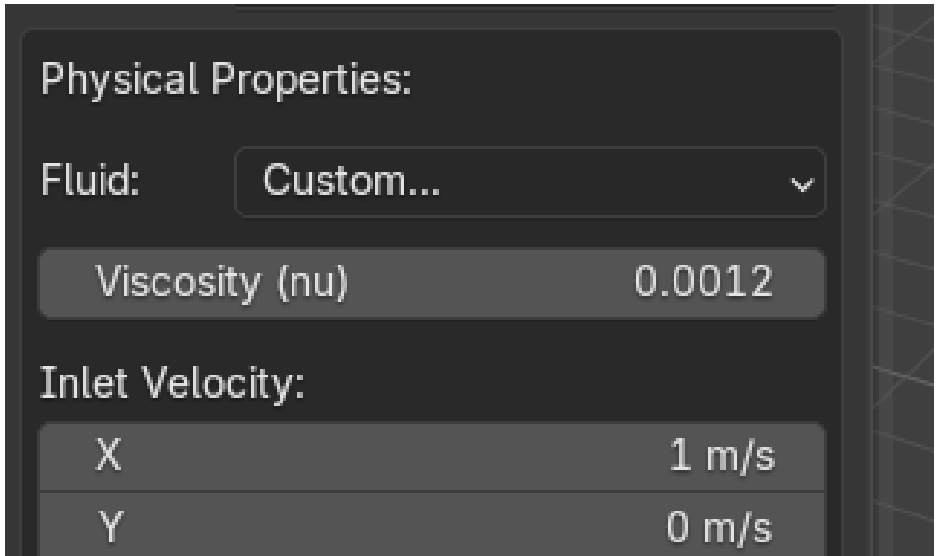
- Constructed a dropdown preset menu mapping predefined real-world constraints:
  - **Air**: $\nu = 1.5 \times 10^{-5}$
  - **Water**: $\nu = 1.0 \times 10^{-6}$
  - **Engine Oil**: $\nu = 1.0 \times 10^{-4}$
  - **Glycerin**: $\nu = 1.18 \times 10^{-3}$
- Directly interfaces with the internal Turbulence & Y+ Calculus mechanisms.



*Fluid Presets Setting

3. Smart UI Interactivity & "Custom" Mode

- **Objective**: Prevent users from accidentally modifying the physical properties of a predefined template, while enabling complete freedom.
- **Dynamic Grey-Out UI Property Box**:
  - Whenever a standard liquid (e.g. Water) is chosen, its viscosity property is visibly populated but explicitly *greyed out* (disabled edit functionality).
  - Selecting the **Custom...** preset removes the greyed-out visual lock, freeing the viscosity input value constraint to float arbitrarily.



4. Timeline & Milestones

Week	Task	Status
1	Project Selection & Initial Briefing	Done
2	OpenFOAM setup, cavity tutorial & ParaView workflows	Done
3	Multi-solver backend, Async non-blocking execution	Done
4	Turbulence models (RANS), Mega exception handling	Done
5 – 6	STL Importer, Jet Colormaps, Residuals & Y+ Calculus	Done
7	System file opener, Custom liquid presets & Smart UI	Done
7	Auto STL repair mechanism	Pending
8 – 12	Real-world geometry benchmarks, Documentation & Demo	Upcoming

Progress Summary (Week 7)

Feature/Milestone	Outcome/Benefit
Notes Migration	Replaced raw unstructured notes into clean task lists
Native File Opener	Opens directories inside Linux file browser instantly
Extended Liquid Dropdown	Native mappings covering Air, Water, Oil, and Glycerin
Dynamic GUI Interactivity	Locks out standard fields. Enabled "Custom" logic.
Auto-Updating Turbulence	Values recursively react off Enum property callbacks.
Version Packaging	Upgraded Blender Source Addon packaged as `cfmesh_tools_2.5.zip`

Thank You!

End of Week 7 Report